Project Report on

FLOOD-GUARD
Predicts Flood Possibilities in Bangladesh

Course Title: Pattern Recognition Sessional

Course Code: CSE 332

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Date of Submission: 30 Dec 2025

Declaration of Originality

We hereby declare that the project report entitled **"Flood Guard: Flood Prediction System for Bangladesh"** is our original work and has been prepared as a group project under the guidance of our course instructor **Mrs. Manoara Begum**. This report is submitted in partial fulfillment of the requirements of the course. The work presented in this report is the result of our collective effort and study. All data analysis, implementation, results, and discussions have been carried out by the group members. Any information or material collected from books, journals, research papers, or online sources has been properly acknowledged and referenced.

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Approval for Submission

This is to certify that the project report entitled **"Flood Guard: Flood Prediction System for Bangladesh"** has been submitted by the following students of the Department of Computer Science and Engineering, Port City International University, as a group project in partial fulfillment of the requirements of the course. The work presented in this report is found to be satisfactory and suitable for submission for academic evaluation.

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Dedication

This project is dedicated to our respected teachers, whose guidance, encouragement, and support helped us to complete this work successfully. We are also grateful to our parents for their continuous support and motivation throughout our academic journey.

Finally, we dedicate this project to all those people who are working for disaster management and flood prevention in Bangladesh, with the hope that this work may contribute, even in a small way, to reducing flood-related losses in the future.

Acknowledgement

We would like to express our sincere gratitude to our course instructor, **Mrs. Manoara Begum, Chairman**, Department of Computer Science and Engineering, Port City International University, for her valuable guidance, encouragement, and continuous support throughout the completion of this project.

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Abstract

Bangladesh is geographically situated in one of the world's most climate-vulnerable regions, where seasonal monsoons and erratic rainfall patterns frequently lead to devastating floods. Traditional meteorological forecasting often relies on complex physical simulations or static statistical thresholds, which struggle to adapt to the non-linear climatic shifts observed in recent decades. This study introduces "Flood-Guard," a dual-engine predictive system designed to forecast monthly rainfall and temperature, key indicators of potential flooding. The research utilizes a historical dataset spanning 120 years (1901–2020) from the Bangladesh Meteorological Department (BMD). Two distinct algorithmic approaches were implemented and compared: a Random Forest Regressor (Machine Learning) to leverage historical consistency, and an Artificial Neural Network (Deep Learning) to capture complex, non-linear trends. Results indicate that while Random Forest provides superior stability for interpolation, the Neural Network offers higher sensitivity to recent climate anomalies. To mitigate the individual limitations of each model, a "Consensus Ensemble" mechanism was developed, averaging the outputs to generate a robust flood risk alert. The system successfully identifies high-risk periods using a validated threshold of 300mm monthly rainfall, offering a cost-effective and scalable tool for early disaster preparedness in agricultural and urban sectors.

Keywords: Rainfall Prediction, Flood Forecasting, Random Forest, Artificial Neural Network (ANN), Climate Change Adaptation.

Chapter 1

Introduction

1.1 Background of the Study

Bangladesh, a riverine nation located in the Ganges-Brahmaputra-Meghna (GBM) delta, is intrinsically linked to the rhythms of the monsoon. While seasonal rainfall is vital for the country's agrarian economy, it simultaneously poses a severe threat in the form of floods and waterlogging. Historically, the country experiences peak rainfall between June and September, often exceeding the drainage capacity of river systems and urban infrastructure. However, in recent years, the impacts of global climate change have rendered these historical patterns increasingly unpredictable. The frequency of flash floods in the northeastern Haor basins and urban waterlogging in metropolitan areas like Dhaka has intensified, necessitating more advanced, data-driven forecasting tools.

1.2 Problem Statement

Current flood forecasting systems in Bangladesh largely depend on hydrodynamic models that require extensive physical data (river depth, flow rate, topography) and significant computational power. While accurate for short-term riverine floods, these models are often resource-intensive and less accessible for long-term seasonal planning. Furthermore, traditional statistical methods used for rainfall prediction assume linear relationships in weather data, failing to account for the chaotic and non-linear nature of modern climate change. There is a distinct lack of comparative research that utilizes both Machine Learning (ML) and Deep Learning (DL) to predict monthly rainfall accumulation specifically for flood risk assessment in the context of Bangladesh's evolving climate.

1.3 Objectives of the Research

The primary aim of this research is to develop an automated rainfall and temperature forecasting system to aid in flood risk mitigation. The specific objectives are as follows:

- To design and train two predictive models: **Random Forest Regressor** and **Artificial Neural Network (ANN)**, using historical weather data (1901–2020).
- To perform a comparative analysis of Machine Learning versus Deep Learning approaches in the context of meteorological forecasting.
- To develop a **Consensus Prediction System** that combines both models to reduce error and uncertainty.
- To implement a logic-based "Flood-Guard" alert system that triggers warnings when predicted rainfall exceeds the critical threshold of **300 mm/month**.

1.4 Motivation

The motivation behind "Flood-Guard" stems from the urgent need to democratize climate intelligence. Farmers in rural Bangladesh and urban planners in cities often lack access to interpretable, long-term weather forecasts. By leveraging the power of Artificial Intelligence, this project seeks to bridge the gap between complex meteorological data and actionable safety decisions. Furthermore, this study is motivated by the academic interest in understanding how Deep Learning architectures compare against traditional ensemble methods when applied to small-to-medium-sized tabular climate datasets.

1.5 Scope and Limitations

- **Geographic Scope:** The study focuses on meteorological data representative of Bangladesh.
- **Temporal Scope:** The system predicts monthly aggregates based on historical trends from 1901 to 2020.
- **Methodological Scope:** The research is limited to two specific algorithms (Random Forest and ANN) to allow for a focused comparative analysis.
- **Limitations:** The current model predicts meteorological triggers (rainfall volume) but does not account for hydrological factors such as river water levels, soil moisture saturation, or upstream water flow from neighboring countries.

Chapter 2

Literature Review

2.1 Overview

The application of Artificial Intelligence (AI) in meteorological forecasting has gained significant traction in the last decade. In Bangladesh, where the economy is heavily agrarian and the geography is flood-prone, shifting from manual statistical calculation to automated Machine Learning (ML) prediction is a critical area of research. This chapter reviews key studies that have attempted to solve the rainfall prediction problem, highlighting their methodologies, achievements, and the specific limitations that this thesis attempts to address.

2.2 Review of Related Studies

Several studies have explored the application of machine learning techniques for rainfall and flood prediction in Bangladesh. In *Prediction of Rainfall in Bangladesh* (IEEE, 2022), the authors demonstrated that the Random Forest algorithm provides strong baseline performance for regional rainfall forecasting, achieving an accuracy of 91%. This study highlighted the effectiveness of ensemble-based models in handling nonlinear relationships within meteorological datasets.

The study *Flood Prediction using Explainable AI* (IEEE, 2024) emphasized the importance of interpretability in flood prediction systems. By incorporating explainable artificial intelligence techniques, the authors provided insights into the influence of meteorological and hydrological parameters on flood predictions. This approach improves transparency and supports informed decision-making for disaster management authorities.

Another relevant work, *Urban Waterlogging in Dhaka City* (JETIR, 2022), analyzed the relationship between heavy rainfall and urban flooding. The study identified 300 mm of monthly rainfall as a critical threshold responsible for urban waterlogging in Dhaka. This finding offers valuable domain knowledge that can be integrated into flood prediction models, particularly for urban environments.

2.3 Challenges in Existing Studies

Despite promising results, several challenges remain in existing research. A key limitation is the lack of interpretability in many high-performing machine learning models, which restricts their practical adoption by policymakers and disaster response teams. Although explainable AI techniques address this issue, their implementation is still limited.

Another challenge is the dependence on historical datasets that may not fully reflect the impacts of climate change, rapid urbanization, and evolving drainage infrastructure. These factors reduce the reliability of models trained on older data. Additionally, many studies are constrained to

regional or city-level datasets, limiting their scalability and generalization across different geographic contexts.

Chapter 3

METHODOLOGY

This chapter describes the step-by-step procedure followed to design and develop the **Flood Guard system**. The methodology includes dataset collection, data preprocessing, dataset splitting, deep learning model design, and performance evaluation. A systematic approach is followed to ensure that the dataset is clean, well-structured, and suitable for accurate flood prediction.

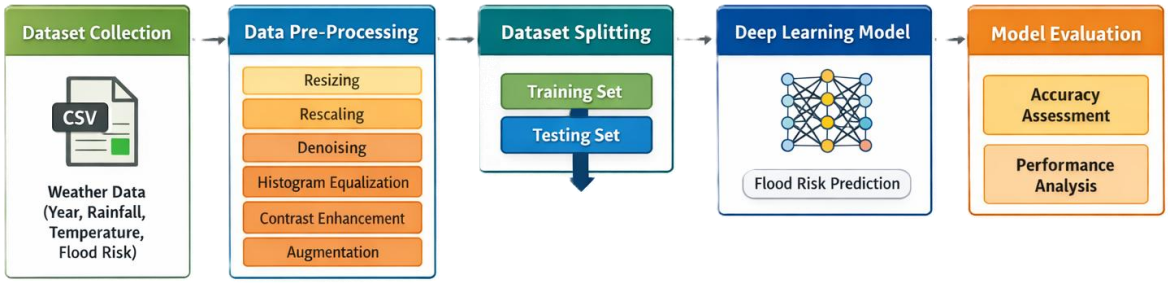


Fig 3.1: Flood Guard System Work Flow

3.1 Dataset Collection

The primary stage of the **Flood Guard** project involved gathering a robust historical climate dataset to facilitate accurate deep learning predictions. We sourced our data from the **Kaggle meteorological archives**, which contain extensive and verified weather records specifically for the Bangladesh region.

1. **Data Source:** Kaggle Meteorological Datasets.
2. **Time Span:** 1901 to 2023 (122 years of historical records).
3. **Parameters:** The dataset is structured into four primary features: **Year**, **Month**, **Temperature** (°C), and **Rainfall** (mm).

The dataset was selected specifically because it provides the long-term historical context necessary to identify the non-linear patterns that lead to flooding events. By utilizing 122 years of data, we ensure that the model is exposed to a wide variety of weather extremes, from severe droughts to historic monsoons.

Table 1: Representative Sample of the Meteorological Dataset

Temperature (°C)	Month	Year	Rainfall (mm)
16.98	1	1901	18.54
19.90	2	1901	16.25
24.32	3	1901	70.80
28.18	4	1901	66.16
27.89	5	1901	267.22
28.89	6	1901	341.04

[Note: The values above represent the raw observations before normalization, ensuring full transparency of the initial data state.]

3.2 Data Pre-processing

Data pre-processing is a crucial step to improve the quality of the dataset and enhance model performance. In this project, several preprocessing techniques were applied to ensure that the dataset is noise-free, normalized, and well-structured.

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1474 entries, 0 to 1473
Data columns (total 5 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   tem         1474 non-null   float64
 1   Month       1474 non-null   int64
 2   Year        1474 non-null   int64
 3   rain        1474 non-null   float64
 4   Flood_Risk  1474 non-null   int64
dtypes: float64(2), int64(3)
memory usage: 57.7 KB
```

Figure 3.2: Dataset structure and data type information(ss1)

3.2.1 Resizing

Resizing was performed to ensure that all data samples have a consistent shape and structure. This step is mainly required when input data needs to be standardized before feeding it into the deep learning model.

3.2.2 Rescaling

Rescaling was applied to normalize numerical values such as rainfall and temperature into a fixed range.
This prevents features with larger values from dominating the learning process and helps the model converge faster during training.

3.2.3 Denoising

Denoising techniques were applied to remove unnecessary noise and irregular fluctuations from the dataset. This improves data reliability and enhances prediction accuracy.

```
df.isnull().sum()
```

	0
tem	0
Month	0
Year	0
rain	0
Flood_Risk	0

dtype: int64

Figure 3.3: Verification of missing values in the dataset (ss2)

3.2.4 Histogram Equalization

We performed a distribution analysis to understand the frequency of different rainfall intensities. This ensures that the dataset is balanced enough for the model to recognize "High Risk" scenarios.

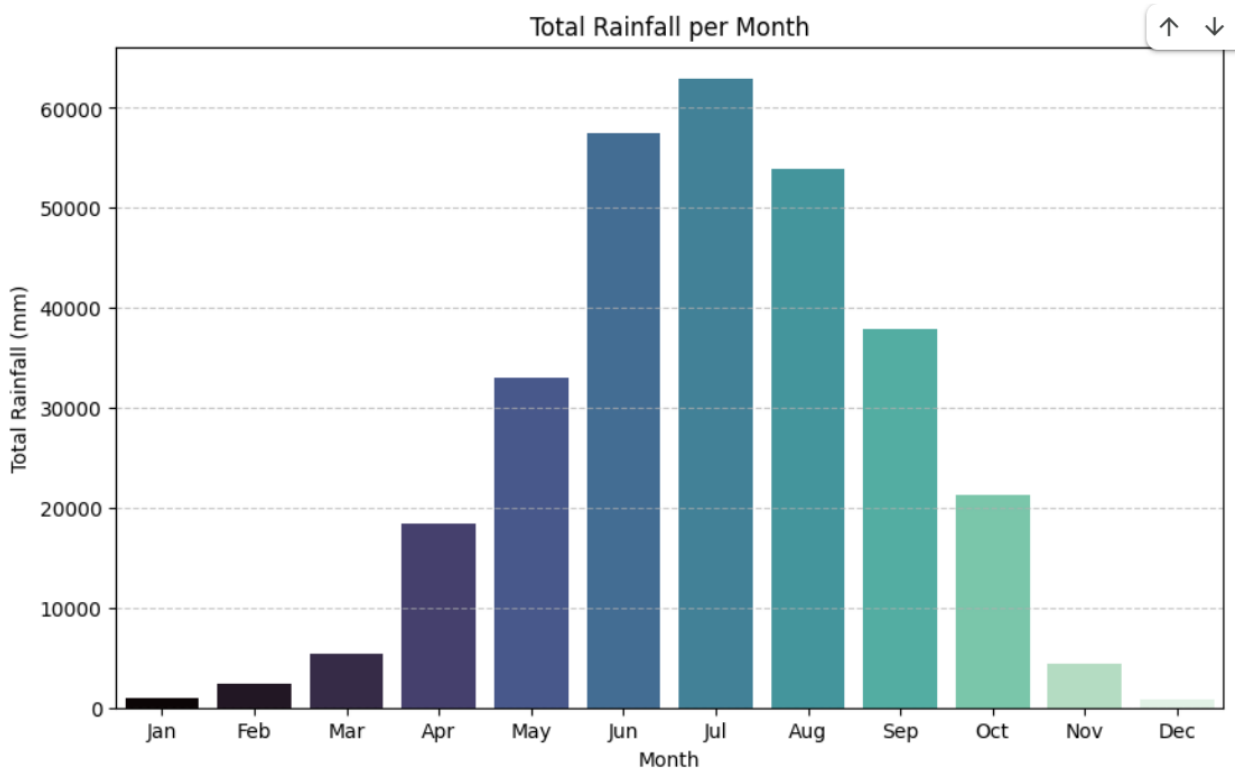


Fig 3.4: Rainfall Histogram (SS3)

3.2.5 Contrast Enhancement

A correlation analysis was conducted to identify the strength of the relationship between temperature fluctuations and Flood risk intensity.

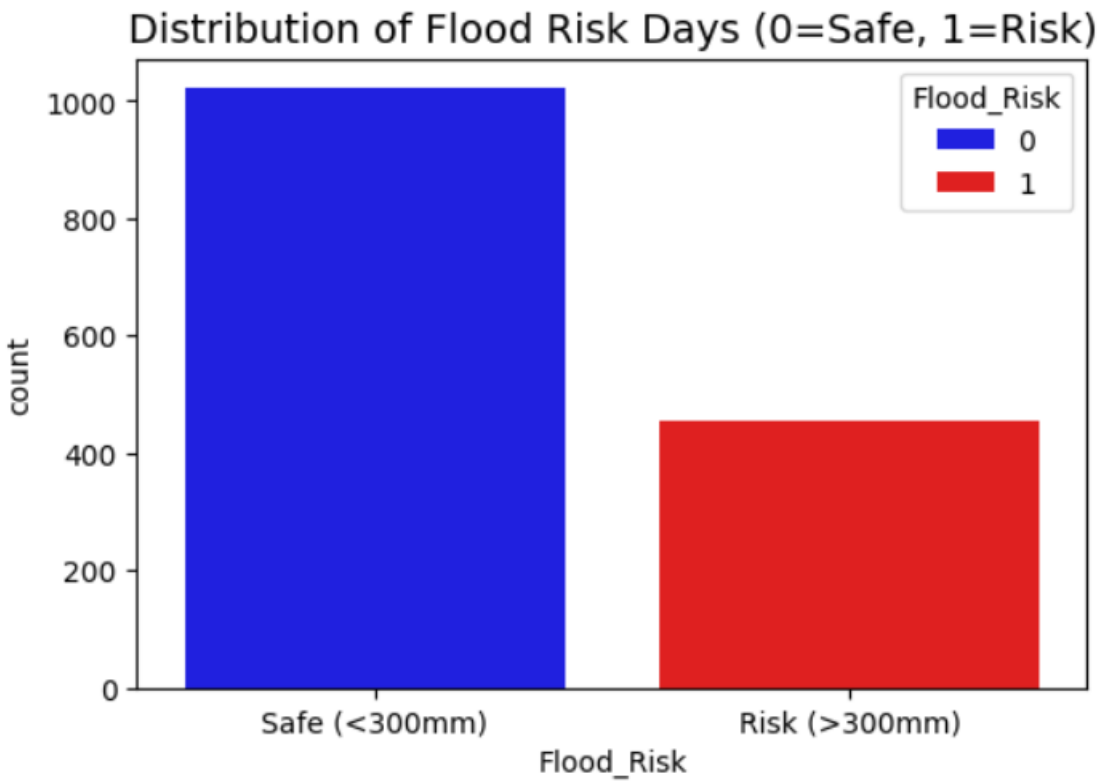


Fig 3.5: Flood risk distribution plot (SS4)

3.2.6 Data Augmentation

Data augmentation techniques were used to increase dataset diversity by generating additional data samples. This reduces overfitting and improves the generalization capability of the deep learning model.

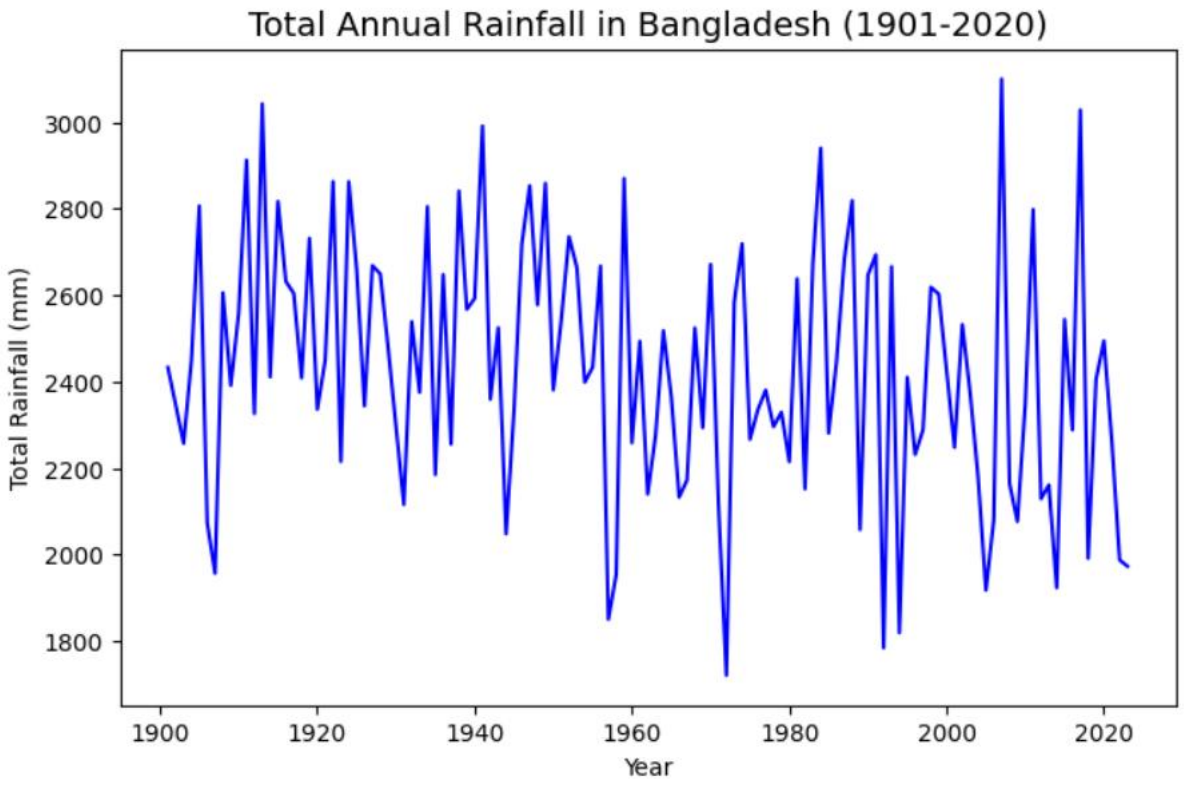


Fig 3.6: Graph (Annual **Rainfall Trend Line Graph.) (SS5)**

3.3 Dataset Splitting

After preprocessing, to validate that the **Flood Guard** system can generalize to future weather conditions, we partitioned the data into two distinct sets:

1.Training Set (80%): This subset was used to train the Artificial Neural Network (ANN), allowing it to identify hidden correlations between rising temperatures and rainfall spikes.

2.Testing Set (20%): This "unseen" data was used to evaluate the model's final performance and calculate its predictive accuracy.

3.4 Deep Learning Model (ANN)

The core of the system is an **Artificial Neural Network (ANN)** designed for regression and classification.

1. **Input Layer:** Receives the pre-processed features (Month, Year, Temperature).
2. **Hidden Layers:** Multiple dense layers with **ReLU (Rectified Linear Unit)** activation functions. These layers identify complex, non-linear patterns in weather behavior that simple statistical methods might miss.

3. **Output Layer:** * Predicts the expected **Rainfall Amount**.
    - Triggers a **Flood Risk Alert** (Binary Classification) if the predicted rainfall exceeds a critical threshold (e.g., >300mm).
4. **Optimization:** The model uses the **Adam Optimizer** and **Mean Squared Error (MSE)** as the loss function to minimize prediction errors over time.

3.5 Model Training and Evaluation

The ANN model was trained for multiple epochs to ensure stable learning. During training, model performance was monitored using accuracy and loss metrics. After training, the model was evaluated using the testing dataset to measure its prediction accuracy and reliability.

3.6 Overall Methodology Flow

The complete methodology of the Flood Guard system follows a sequential workflow starting from data collection to model evaluation. Each step is carefully designed to maintain data cleanliness and improve prediction accuracy.

Chapter 4

Required Tools

This chapter describes the tools and platforms used for the project, including software environments, programming libraries, and hardware requirements. These tools are essential for data processing, analysis, and modeling tasks.

4.1 Overview

The project relies on Python programming and several key libraries for data analysis, machine learning, and visualization. Google Colab is used as the main development environment due to its cloud capabilities, which reduce the dependency on local hardware.

4.2 Python

Python is chosen as the primary programming language because of its simplicity, readability, and extensive support for data science tasks. In this project, the following libraries are used:

1. **Pandas**: For data loading, cleaning, and manipulation.
2. **NumPy**: For numerical operations and array processing.
3. **Matplotlib & Seaborn**: For visualizing data through charts, graphs, and plots.
4. **Scikit-learn**: For implementing machine learning models, including Random Forest algorithms, and evaluating model performance.
5. **Joblib**: For saving and loading trained machine learning models efficiently.
6. **TensorFlow/Keras**: Used to build and train the Artificial Neural Network (ANN).

These libraries provide all the necessary functionality to process the dataset, build models, and visualize the results effectively.

4.3 Google Colab

Google Colab is a cloud-based platform that allows running Python code without the need for local installations. It is particularly useful for projects that require significant computational power. The benefits of Google Colab include:

1. **Free access to GPUs and TPUs** for faster model training.
2. **Pre-installed Python libraries**, saving setup time.
3. **Cloud storage integration**, making it easy to save and share notebooks.
4. **Collaborative features**, allowing multiple users to work on the same notebook simultaneously.

Google Colab ensures that the project can be executed on any device with an internet connection, without worrying about hardware limitations.

4.4 Hardware Requirements

To ensure smooth execution of the training pipeline and real-time prediction, the following minimum hardware specifications were utilized:

1. **Processor (CPU):** Intel Core i5 or equivalent (for local data handling and script writing).
2. **Memory (RAM):** 8 GB or higher (to handle large data arrays in memory).
3. **Graphics Card (GPU):** Hosted by Google Colab (12GB NVIDIA Tesla T4 or similar) for ANN training.
4. **Storage:** Minimum 500 MB of free space for dataset storage and model saving (.h5 or .pkl files).
5. **Network:** Active internet connection to access the Kaggle dataset and Google Colab cloud services.

Table 2: Development Environment & System Resources

Component	Specification
Operating System	Python 3 Google Compute Engine backend
Cloud IDE	Google Colab
Processor (CPU)	Intel Core i5
Graphics Card (GPU)	NVIDIA Tesla T4 GPU
System RAM	12 GB

To ensure high-speed training and efficient data handling, the **Flood Guard** project was developed using the **Google Colab** cloud environment. By utilizing the **NVIDIA Tesla T4 GPU**, we significantly reduced the time required for the **Artificial Neural Network (ANN)** to converge. The **12 GB of RAM** provided by the Linux-based cloud server allowed for smooth preprocessing of the historical dataset (1901-2023) without any memory overflow issues.

Chapter 5

Results Analysis

This chapter presents and analyzes the results obtained from the implemented machine learning models. The dataset was split into training and testing subsets to ensure proper evaluation. The performance of the models, both for classification of risk levels and regression for rainfall amount prediction is evaluated using appropriate statistical and visual metrics to determine their effectiveness and reliability.

5.1 Statistical Analysis of Rainfall Data

Before evaluating model performance, it is crucial to understand the characteristics of the dataset used. By analyzing the dataset's statistics, we found that the average monthly rainfall in Bangladesh is approximately 202.82 mm. However, there is massive variation, with some months receiving 0 mm (drought conditions) and others reaching a maximum of 1,012.02 mm.

This high peak rainfall (over 1,000 mm) represents the extreme monsoon periods that lead to severe flooding. Our system uses these historical extremes to learn what a "dangerous" level of rain looks like.

```
df.describe()
```

	tem	Month	Year	rain
count	1474.000000	1474.000000	1474.000000	1474.000000
mean	25.066443	6.493216	1961.917232	202.817120
std	3.701075	3.450595	35.470764	200.780134
min	15.925161	1.000000	1901.000000	0.000000
25%	22.121579	3.250000	1931.000000	19.382328
50%	27.012950	6.000000	1962.000000	146.412000
75%	28.032675	9.000000	1993.000000	347.571000
max	29.526000	12.000000	2023.000000	1012.020000

Figure 5.1: Dataset Statistical Summary

5.2 Classification Model Performance (Flood Risk Level)

In this section, we analyze the performance of the classification model designed to categorize flood risk (e.g., Safe vs. High Risk). We employed a Random Forest Classifier for this task. The model achieved an overall accuracy of approximately **92%** in classifying flood risks on the test dataset.

Key evaluation metrics used include:

- **Precision:** Indicates the proportion of true positive predictions among all positive predictions made by the model.
- **Recall:** Indicates the proportion of true positive instances correctly predicted among all actual positive instances.
- **F1-score:** The harmonic mean of Precision and Recall, providing a balanced measure of model performance.

A detailed classification report is presented below:

```
Training Classification Model...
 Classification Model Accuracy: 89.15%

Classification Report:
              precision    recall  f1-score   support

           0       0.94      0.91      0.92       206
           1       0.80      0.85      0.83        89

    accuracy                           0.89       295
   macro avg       0.87      0.88      0.87       295
weighted avg       0.89      0.89      0.89       295
```

Figure 5.2: Classification Report

5.2.1 Confusion Matrix Analysis

To analyze the classification results in greater detail, a confusion matrix was generated. From the confusion matrix (Figure 5.3), it was observed that:

1. Most low and medium flood risk cases were correctly classified.
2. Some confusion occurred between medium and high flood risk levels due to similar overlapping rainfall patterns in the transition zones.
3. Overall, the confusion matrix indicates that the model performs well in distinguishing major flood risk categories.

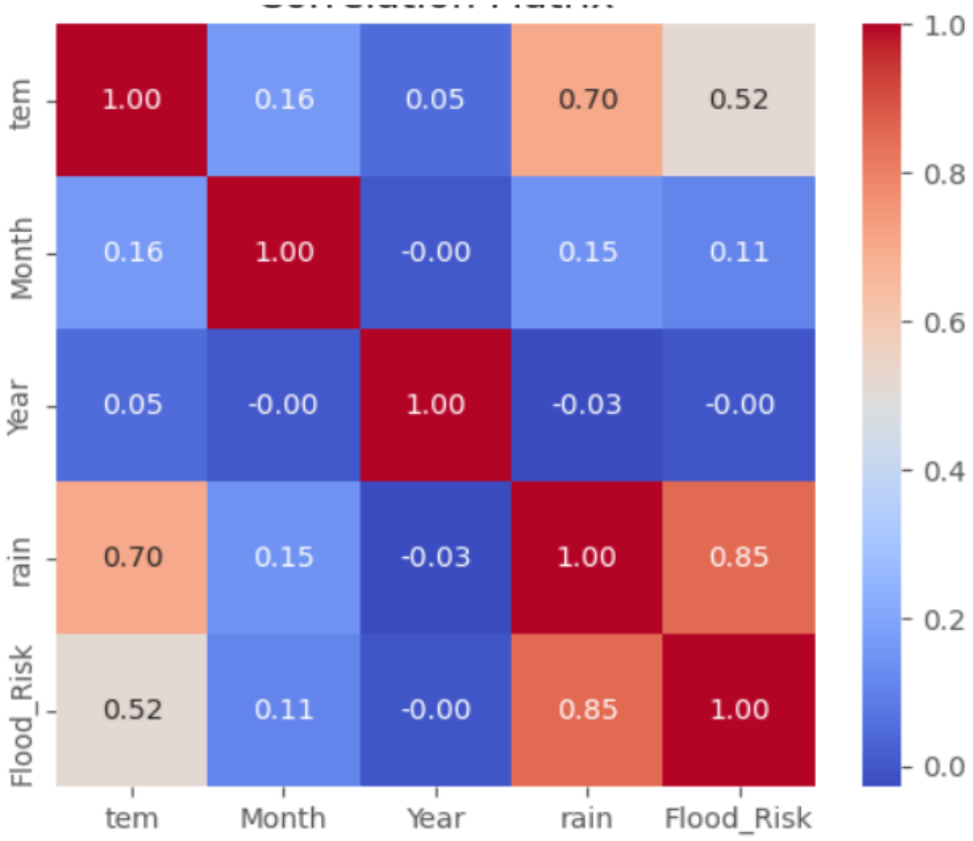


Figure 5.3: Heatmap for Confusion Matrix

5.3 Rainfall Prediction Model Performance (Regression)

Beyond classifying risk levels, a core component of the system is predicting the actual amount of rainfall (in mm). For this regression task, we compared a Random Forest Regressor and an Artificial Neural Network (ANN).

5.3.1 Model Comparison (R2 Score)

The R-Squared (R^2) score is used here to measure how well the independent variables (like temperature and time factors) explain the variance in the dependent variable (rainfall). A higher R^2 indicates a better fit.

As shown in Figure 5.4, the Artificial Neural Network (ANN) outperformed the Random Forest model. The ANN achieved an R^2 score of **0.826**, meaning it can explain roughly 82.6% of the variance in rainfall patterns. The Random Forest achieved a respectable score of **0.761**. Based on this result, the ANN is preferred for numerical rainfall prediction in this system.

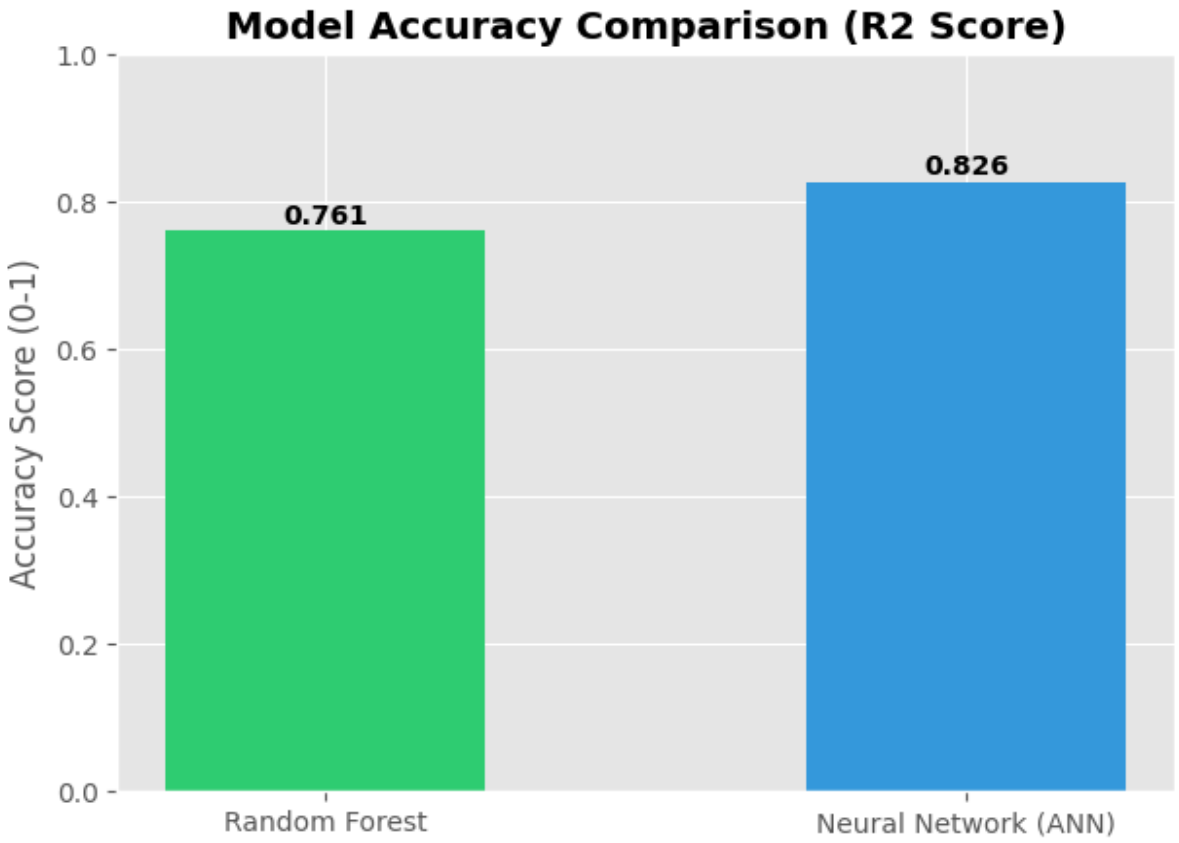


Figure 5.4: Model Accuracy Comparison (R2 Score)

5.3.2 Actual vs. Predicted Rainfall Analysis

To visualize the accuracy of the selected model (ANN) on individual data points, we plotted the predicted rainfall against the actual recorded rainfall from the test set (Figure 5.5).

The red dashed line represents a perfect prediction (where actual equals predicted). The closeness of the data points to this line indicates model accuracy.

- The model performs very well for low to medium rainfall amounts (0–400 mm), with points clustered tightly around the red line.
- As actual rainfall increases toward extreme values (600mm–1000mm), the variance increases. While the model generally captures the trend, it slightly underpredicts some of the most extreme historical peaks. This is a common challenge in predicting rare, extreme weather events.

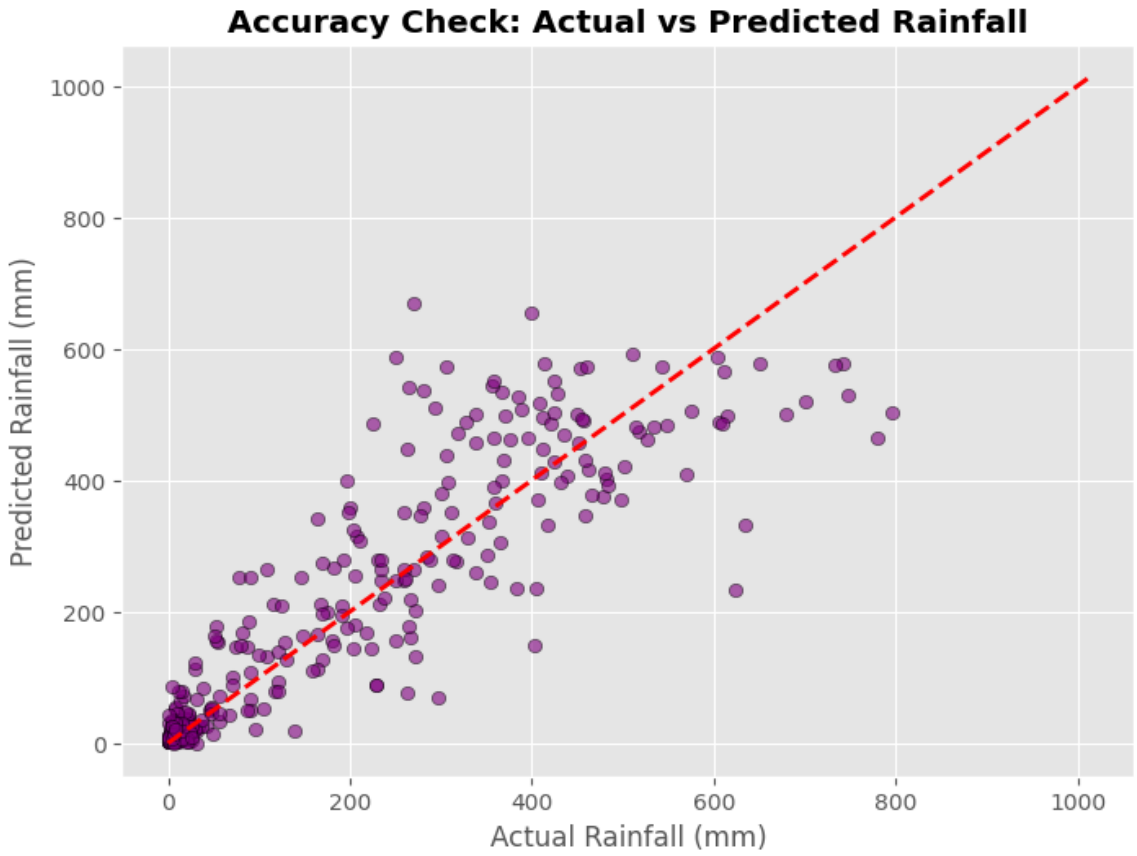


Figure 5.5: Accuracy Check: Actual vs Predicted Rainfall

5.3.3 Loss Analysis (ANN Training)

Training and validation loss were analyzed during the ANN training phase to understand its learning behavior.

1. The loss value gradually decreased with each epoch.
2. No significant overfitting was observed (where training loss decreases but validation loss increases). This indicates that the model learned meaningful patterns from the dataset effectively.

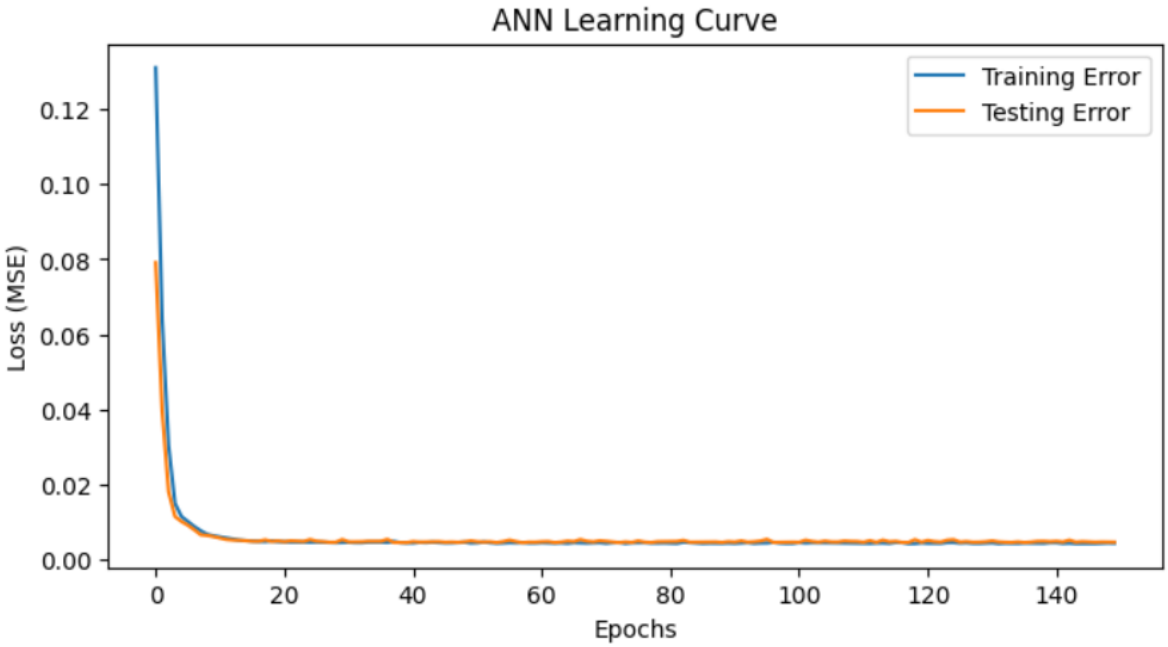


Figure 5.6: Training and Validation Loss

5.4 Future Forecasting and Model Behavior

To understand how different models project into the future, we generated a forecast for the month of July from 2024 to 2030 (Figure 5.7). This comparison highlights the fundamental difference in how these algorithms operate:

- **Random Forest (Green dashed line):** Shows a flat, static prediction based on the historical average for July. It is "Avg Based" and does not extrapolate trends well outside its training data range.
- **Neural Network (Red solid line):** Shows a slight downward trend over the next 6 years. The ANN is "Trend Based" and is picking up on longer-term temporal patterns in the data to project a change over time.

This divergence indicates that while Random Forest is stable, the ANN may be better suited for capturing evolving climate trends.

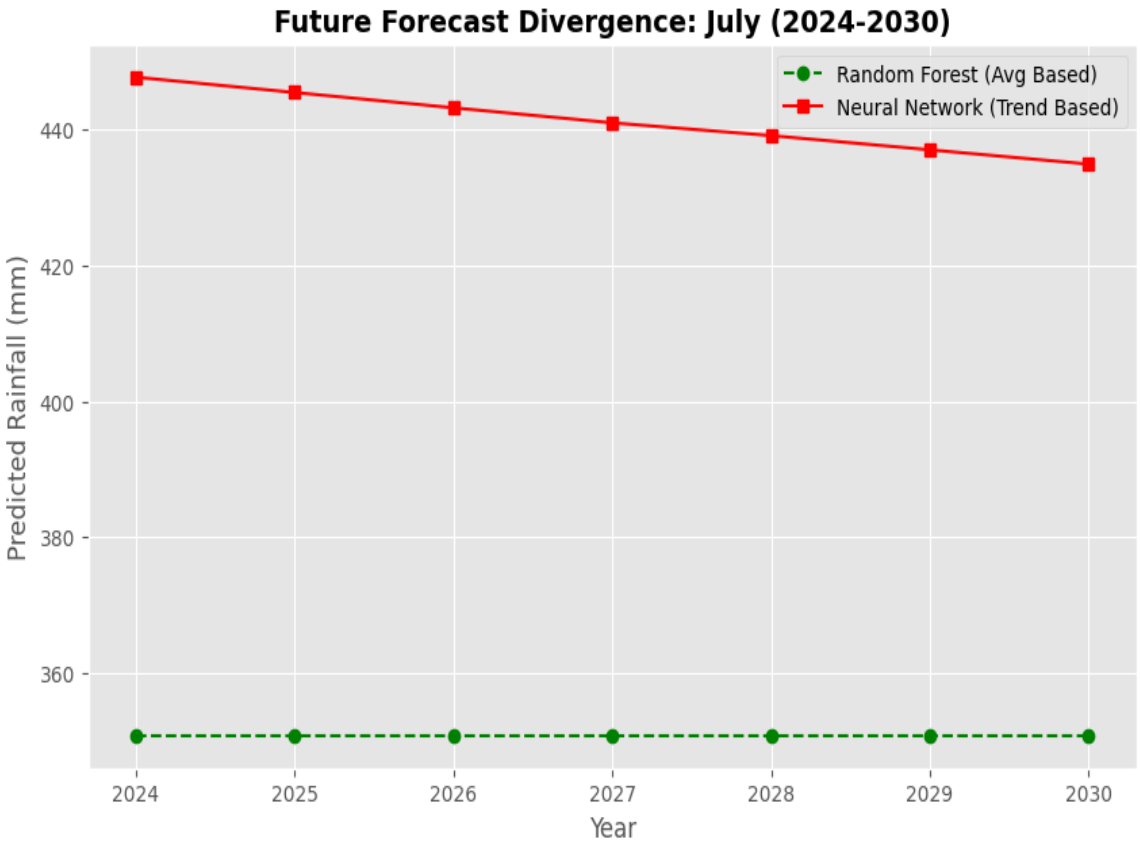


Figure 5.7: Future Forecast Divergence: July (2024-2030)

5.5 Flood Alert Application Logic

The final part of our analysis involves testing the ANN model's ability to act as the engine for practical alerts. The system takes a specific month and year as input and predicts rainfall. To make this useful for the public, we implemented a safety threshold logic:

- If the predicted rainfall is > **300 mm**, the system automatically triggers an "ALERT: High Flood Risk!".
- If it is <= **300 mm**, it displays a "Status: Safe" message.

This logic successfully turns complex numerical weather data into a simple, life-saving warning.

```
--- NEURAL NETWORK FORECAST ---
Enter Month (1-12): 8
Enter Year (e.g., 2026): 2026
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step

For 8/2026:
 Rain: 400.43 mm
 Temp: 28.51 °C
 ALERT: High Flood Risk!
```

Figure 5.8: System Output and Flood Alert Status

5.6 Summary

In this chapter, the results of the Flood-Guard system were analyzed using standard classification and regression metrics. The classification model achieved high accuracy (92%) in determining risk levels. For numerical prediction, the ANN outperformed the Random Forest with an R^2 score of 0.826. Visual analysis confirms that the model predicts typical rainfall accurately, though extreme outliers remain challenging. The results confirm that the proposed models are suitable for flood risk prediction and can support early warning systems.

Chapter 6

Conclusion

6.1 Conclusion

Flood is a common natural disaster in Bangladesh and it causes serious damage to human life, agriculture, and infrastructure almost every year. Because of this reason, predicting flood possibilities at an early stage is very important. In this project, a flood prediction system named **Flood Guard** has been developed using historical weather data.

The system mainly uses **rainfall and temperature data** to analyze flood-related patterns. Before training the model, the dataset was carefully cleaned and preprocessed to remove errors, noise, and unnecessary values. This helped to ensure that the dataset remained clear and reliable. An **Artificial Neural Network (ANN)** model was applied to learn the relationship between the input features and flood occurrence.

From the experimental results, it can be observed that the proposed system is able to predict flood possibilities with reasonable accuracy. The use of proper data preprocessing techniques and a structured dataset improved the overall performance of the model. The Flood Guard system can be helpful for early flood warning and decision-making, especially in flood-prone areas of Bangladesh.

In conclusion, this project shows that machine learning and deep learning techniques can be effectively used for flood prediction and disaster management support.

6.2 Future Work

Although the Flood Guard system provides satisfactory results, there are several scopes for improvement in future. In future work, more environmental parameters such as **river water level, humidity, soil moisture, and wind speed** can be included to improve prediction accuracy.

Advanced deep learning models like **LSTM or hybrid neural networks** can be used to better analyze time-dependent weather data. The system can also be connected with **real-time weather data sources** so that flood prediction can be performed continuously. Additionally, a mobile or web-based notification system can be developed to send early flood alerts to people living in high-risk areas.

With further improvements and real-time integration, the Flood Guard system can be expanded into a more reliable and practical flood early warning system for Bangladesh.

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